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Innovative H_2S Management in Sour Wells: Safe Rigless Approach

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Abstract

This study explores the application of a completion design solution which enables downhole sweet gas injection via an insert string tailored for high hydrogen sulfide (H_2S) wells, aimed at managing H_2S concentrations within the wellbore through downhole dilution with an external sweet gas source. This approach enables the controlled reduction of H_2S levels directly through the production path through the wellbore to the surface, allowing hydrocarbon production with acceptable H_2S concentrations. This approach reduces the need for extensive surface processing and enhances the profitability and operational viability of high-H_2S wells.

This abstract presents an innovative, yet simple completion design that enables sweet gas to be injected into the wellbore with minimum required surface injection pressure to get mixed downhole with sour gas to reduce H_2S content by dilution. The system employs advanced downhole engineering to achieve targeted gas injection within high-H_2S wells, allowing precise control of injection depths without the need for rig-based intervention. The installation begins with rigorous well preparation, along with integrity checks, after which an intermediate spool assembly is installed just below the wellhead and secured by a shallow bridge plug. This provides a structurally stable base and pressure isolation for the subsequent insert string or coiled tubing installation. The insert or coiled tubing string, specially equipped with a precision injection valve, is run downhole to achieve accurate gas placement at the designated shallow injection depth. This configuration facilitates uninterrupted downhole gas injection, ensuring controlled H_2S dilution with optimized injection precision to stabilize production flow while maintaining well pressure.

The proposed system demonstrated substantial improvements in H_2S management, economic performance, and operational sustainability. By introducing an external sweet gas source directly into the production stream, the system effectively diluted downhole H_2S concentrations to manageable levels, thus leading to a reduction in the H_2S level in the well-stream. The system also minimizes the need for surface-level treatment yielding reduced downstream processing costs. This method led to significant operational savings by maintaining well pressure stability and achieving efficient fluid lifting without additional mechanical interventions. Moreover, the possible rigless design facilitated faster well restarts following shutdowns, reducing downtime and supporting stable production. In terms of business value, this system will reduce operational costs and production risks in high-H_2S wells. By mitigating surface H_2S processing,

the approach simplifies facility requirements, minimizes environmental and safety risks associated with handling high-H_2S flows, and translates to lower equipment and maintenance expenditures related to the subsurface. The field data supports the conclusion that subsurface dilution enables safe well operations, by reducing the rupture exposure radius (RER). This deleted section is not necessary. This technology will help to unlock the sour gas potential at reduced costs. The rigless and remotely operated system allows wells to be quickly restarted, enhancing production continuity, reducing non-productive time (NPT), and extending well lifespan. The flexibility of the system also permits dynamic responses to fluctuating H_2S levels without costly interventions, securing long-term cost savings and increased profitability.

The rigless downhole gas sweetening using insert string e.g. inverse gas lift system presents a breakthrough in high-H_2S well management by using precision downhole gas dilution to optimize production and reduce processing costs. Its rigless, versatile design promotes sustainable H_2S management, enhances well productivity, and extends operational efficiency without requiring rig-based intervention. This adaptable, proactive approach to H_2S management supports industry sustainability goals by offering a low-impact, optimized solution for managing high-H_2S reservoirs.

Introduction

Sour wells, characterized by the presence of significant concentrations of H_2S pose substantial challenges to hydrocarbon production. These challenges span multiple dimensions, including corrosion, health and safety risks, environmental concerns, and economic implications. H_2S is highly corrosive to the carbon steel commonly used in wellbore tubular and surface facilities, leading to sulfide stress cracking and general corrosion, which can cause equipment failures and leaks (Chen et al., 2020; Wang et al., 2019). The presence of CO_2 exacerbates this issue, leading to increased corrosion rates. From a health and safety perspective, H_2S is extremely toxic, even at low concentrations, posing a severe risk to workers and nearby communities. Exposure can result in respiratory paralysis and death, necessitating stringent safety measures and monitoring systems. The release of H_2S into the atmosphere contributes to acid rain and poses risks to wildlife and ecosystems.

The presence of hydrogen sulfide may stem from diverse geochemical and operational processes (Aniekan et al., 2023). Contributing factors include high-temperature sulfate-hydrocarbon reactions (thermochemical sulfate reduction). The study describes the sweetening of hydrocarbons produced from sour gas wells. Sour gas includes certain acidic compounds, such as H_2S, that can introduce operational and safety issues due to their corrosive properties and toxicity (Patent US11970927 - Bottom hole Sweetening of Sour Gas Wells).

Conventional Production Methods and Technical Limitations

Traditional approaches to managing sour hydrocarbon production rely on chemical, metallurgical, and process engineering strategies. Chemical mitigation methods, such as the injection of corrosion inhibitors (e.g., imidazolines) and H_2S scavengers (e.g., triazines), face challenges in maintaining efficacy across dynamic downhole conditions. Adsorption inefficiencies, thermal degradation, and incomplete tubular coverage often lead to localized corrosion and sulfide stress cracking, particularly in high-temperature, high-salinity environments (Askari et al., 2021; Marks and Martin, 2007). Furthermore, the disposal of reaction byproducts, such as thiocyanate-laden wastewater from scavenger systems, introduces environmental compliance risks, requiring advanced treatment processes to meet regulatory standards. Recent field trials of novel scavenger chemistries, such as sodium nitrite-based systems, demonstrate improved reaction kinetics but highlight trade-offs in compatibility with produced fluids.

Material selection strategies, including the use of corrosion-resistant alloys (CRAs) such as duplex stainless steels and nickel-based alloys, offer improved resistance to H_2S-induced degradation. However, their deployment is constrained by prohibitive capital costs, particularly for retrofitting existing carbon steel infrastructure. CRAs also exhibit vulnerabilities in extreme sour environments, including hydrogen embrittlement under high H_2S partial pressures. Advanced metallurgical solutions, such as modified martensitic steels with reduced sulfur inclusion content, have shown promise in extending service life in wells with high H_2S concentrations (Badrak, 2018).

Other technical strategies such as sour gas reinjection for EOR purposes (Siddiqui et al., 2013); also surface processing techniques, such as amine gas sweetening and sulfur recovery that require significant capital and operational expenditures (Al-Ghanem et al., 2018). Field case studies emphasize the operational challenges of scaling surface H_2S management systems for reservoirs with variable souring profiles.

Other technological approaches for H_2S management include strategies like sour gas reinjection to enhance oil recovery (Siddiqui et al., 2013); and surface-level solutions such as amine-based gas sweetening and sulfur recovery units. However, these methods often demand substantial upfront investments and ongoing operational costs (Al-Ghanem et al., 2018). Field case studies further underscore the complexity of adapting surface H_2S mitigation infrastructure to reservoirs with fluctuating or unpredictable souring patterns, which complicates system scalability and efficiency. Such variability necessitates flexible engineering designs and real-time monitoring to balance economic feasibility with effective H_2S control.

The shortcomings of traditional approaches emphasize the necessity for cross-functional strategies through advanced engineering design, computational modeling, and data analytics. By focusing on economically and sustainable environmentally optimized technologies to achieve proactive H_2S risk mitigation while maximizing asset profitability across the lifecycle of complex sour reservoirs.

Tech-Driven Approach and Engineering Framework

As the global exploration of gas grows continuously, many fields containing high content of Hydrogen Sulfide (H_2S) have to be developed. Since H_2S occurring at high levels is a life-threatening, corrosive and flammable gas, the exploration and operation of such fields has to be undertaken under very strict safety precautions. This implicates a holistic approach concerning the safety, escape and rescue strategy of the facility. Thus, to mitigate the effect of high concentrated H_2S, a new subsurface diluting mechanism will be introduced to lower the H_2S concentration below surface at safe shallow depth prior to gas production start or reach at surface.

The downhole sweet gas injection represents a promising engineering technology for managing H_2S in sour wells. This approached technique involves injecting sweet gas (natural gas with zero or low H_2S content) directly into the wellbore to dilute the concentration of H_2S in the produced fluids thereby unlocking the gas potential safely and at reduced cost. The system incorporates an innovative, yet simple completion design that enables sweet gas injection to be pumped into the wellbore with minimum required surface injection pressure to facilitate downhole mixing with sour gas to reduce H_2S content by dilution. This method allows the production of sour gas while mitigating H_2S negative impact on downhole and surface equipment. The proposed approach can mitigate corrosion, reduce health and safety risks, and lower the costs associated with surface processing (Patent US11970927 - Bottom hole Sweetening of Sour Gas Wells).

The idea enables sweet gas injection inside production tubing that allows:

- Method of sweet gas injection downhole via insert string (e.g. Coiled Tubing) to minimum desired depth
- Production post mixing/diluting sour reservoir gas via annuli between insert string & production tubing
- Use existing xmas tree with minimum modification to wellhead (additional spool)
- Retriable insert string without rig when required

The inverse gas lift system (IGLS) represents a transformative advancement that is implemented within the artificial lift technology, specifically designed to address challenges in low-pressure, mature, or water-dominated reservoirs (Okorocha et al., 2020) to provide sustained production levels and enhance economic returns (Julian et al., 2014). Unlike conventional gas lift systems, which inject gas into production tubing, the IGLS reverses this pathway by introducing gas into the annular space between the casing and tubing. This completion system originally has been developed as a gas lift technology that represent cost-effective and viable options for liquid loading (liquid/gas application). However, it was not utilized for gas to gas application diluting mechanism in sour wells as explained earlier. The application is for wellbore only to dilute H_2S at shallow depth and not changing sour reservoir composition at reservoir (No Enhanced oil Recovery or sweet gas flooding at rock level). The fluid flow composition through porous media will stay unchanged. The mass of H_2S will also not change as dilution occurring at shallow depth to address the sour gas challenges upstream of the wellhead. The volume of sweet gas used for dilution is one time assuming H_2S treatment plant is available to treat injected volume and recycle for the same. In addition, the completion system utilizes same tree with additional spool with dedicated hanger installed which allows both injection and commingle production at same time (Figure 1).

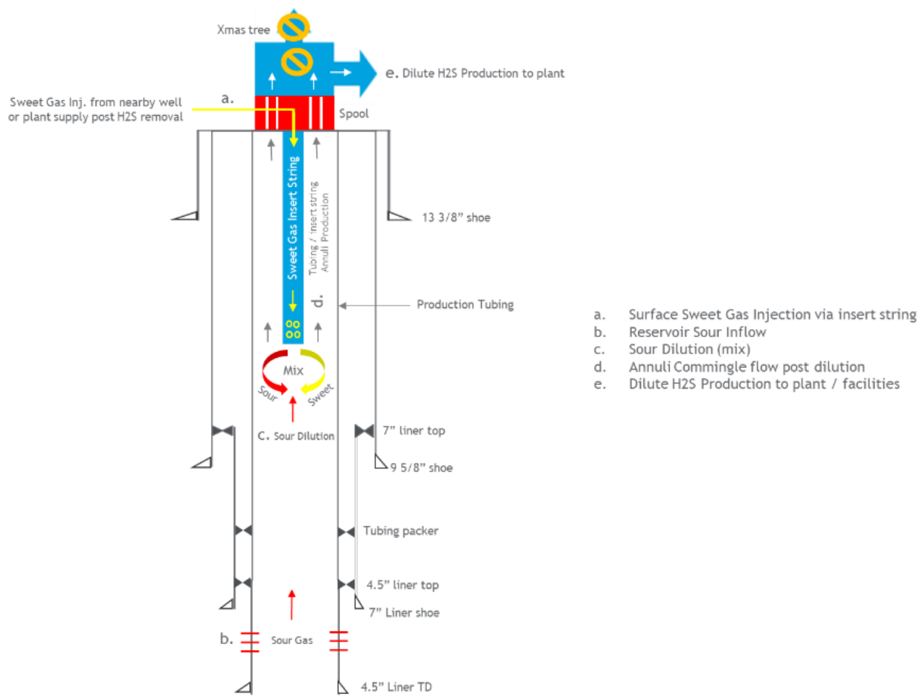

Figure 1—Shallow Sweetening Through Insert String

This rigless approach eliminates the need for high-pressure gas compression and costly rig interventions, enabling precise gas injection at any depth while maintaining operational efficiency and reducing environmental impact (Gamal et al., 2025).

The introduction of novel engineering methodology is a key enabler to deploy the inverse gas lift system as an integrated downhole completion for injecting sweet gas into sour wells as a safe rigless approach. The engineering approach is designed to dilute the native sour gas stream by leveraging controlled gas mixing within the production zone, thereby reducing H_2S concentration to safe operational thresholds through a safe sustainable cost-effective solution.

System Components

The inverse gas lift system is considered an integrated completion architecture comprising upper and lower assemblies designed for rigless gas injection and controlled sour gas dilution. The upper completion integrates an intermediate spool installed below the production Christmas tree, housing a concentric hanger that segregates gas injection and production flow paths. Jointed tubing extends from the wellhead to the subsurface safety valve (SSV) depth, terminating in a seal stinger equipped with a pump-out plug. This stinger engages a polished bore receptacle (PBR) within the lower completion, ensuring a gas-tight connection between the upper and lower assemblies. The lower completion features a dual flow safety valve (DFSV) anchored via a suspension hanger or dual flow hanger (DFH), which secures the assembly while isolating injection and production conduits. Coiled tubing extends from the DFSV to the gas injection valve at depth, enabling precise delivery of sweet gas into the designed injection point to mitigate H_2S concentrations.

The intermediate spool (Figure 2) serves as the primary interface for gas injection, retrofitted below the production tree during IGLS deployment. This component elevates the tree and flowlines to accommodate gas introduction while maintaining well integrity. Internally, the spool incorporates a machined profile to accept the concentric hanger, which bifurcates the injection and production pathways.

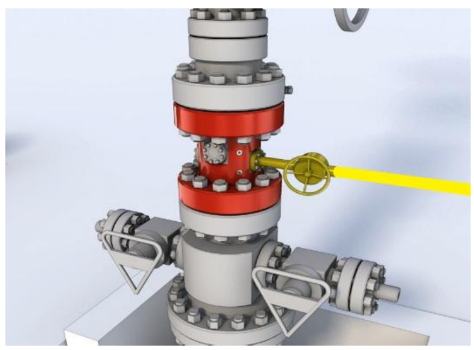

Figure 2—Intermediate Spool Component

The concentric hanger (Figure 3) is central to flow-path segregation. The standard variant, devoid of moving parts, is landed within the intermediate spool and secured via external lockdown bolts. These bolts, installed through 180°-opposed side ports on the spool, engage locking dogs that mechanically fix the hanger to the spool's no-go profile. Before installation, internal seal bolts provide a double-barrier isolation, ensuring integrity during deployment.

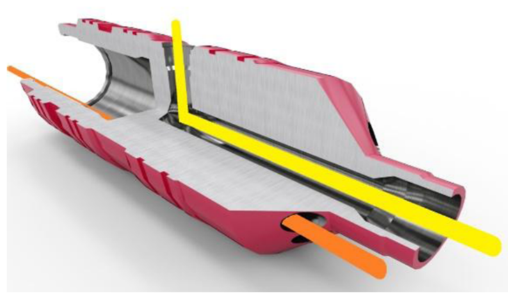

Figure 3—Concentric Hanger

The seal stinger (Figure 4) is run on the bottom of the upper completion tubing, engaging the PBR of the lower completion, and thereby connecting the gas injection conduit from the surface to the injection valve at depth. The lower centralizer is shearable and gives a positive indication of PBR entry on running in the hole prior to full set down, thus allowing tubing space out verification.

Figure 4—Seal Stinger

The dual flow hanger (DFH) integrates with the dual flow safety valve to suspend the lower completion without imposing static loads on the safety valve's no-go profile, which is not rated for long-term weight retention. Injected gas traverses the DFH's central bore, while production fluids ascend through an annular bypass. The PBR, mounted atop the DFH, accepts the seal stinger, completing the hydraulic circuit between upper and lower assemblies.

Figure 5—Dual Flow Hanger

Suspension hangers are deployed in scenarios where existing lock profiles are compromised or unavailable, the suspension hanger employs slip mechanisms to anchor the lower completion. While its central bore facilitates gas injection, production fluids bypass the slips via an annular flow path. Though designed for minimal flow restrictions, the slip assembly may impose diameter limitations, necessitating careful sizing to preserve production rates (Figure 6).

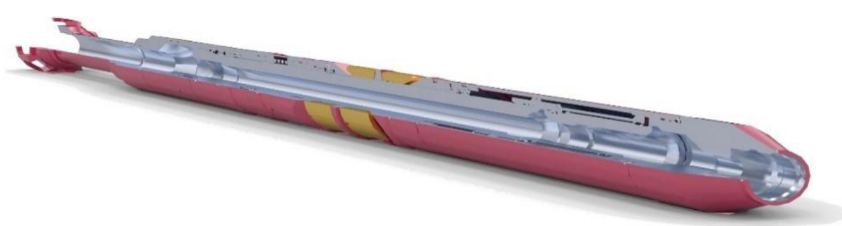

Figure 6—Suspension Hanger

The dual flow safety valve (DFSV) as shown in Figure 7 is the system's linchpin, enabling full well control during gas injection and production. Its concentric flow tube isolates injection and production paths: when open, gas flows through the central bore to the injection valve, while production ascends via the annular channel. Upon closure, the flapper isolates both paths below the valve, with equalization occurring above and below to prevent pressure differentials.

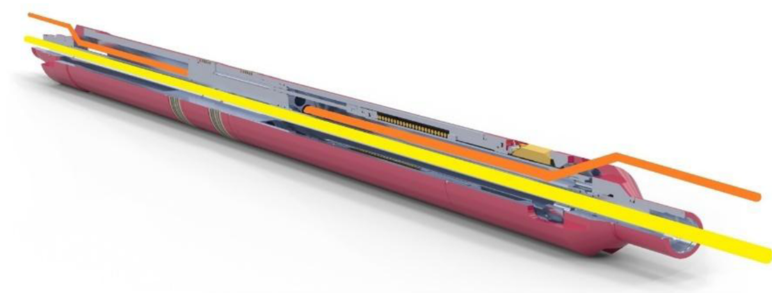

Figure 7—Dual Flow Safety Valve

The gas injection valve is positioned at the target depth via coiled tubing, the gas injection valve regulates sweet gas delivery into the designed injection point. Hydrostatic compatibility ensures optimal deliquification and lift efficiency, with injection rates adjustable to reservoir conditions. The valve's placement flexibility allows customization for varying well geometries and pressure regimes. Coiled tubing deployment utilizes slip connectors for standard applications and grapple-style connectors for high-tension environments to ensure secure attachment of the gas injection valve to the DFSV, withstanding cyclic stress during injection and production cycles.

Gas Dilution Workflow: Simulation and Modeling

This section details the engineering simulation and modeling workflow for managing H_2S in sour well production through gas dilution, as illustrated in Figure 8. The process begins with problem identification, where specific H_2S-related challenges are defined, followed by data analytics to understand well characteristics and H_2S production. A technical requirement assessment then determines the necessary components for an effective gas dilution solution. Next, gas dilution simulation and modeling optimize system performance, leading to system design and subsequent field installation. Finally, continuous monitoring and evaluation ensure the system's ongoing effectiveness in H_2S management.

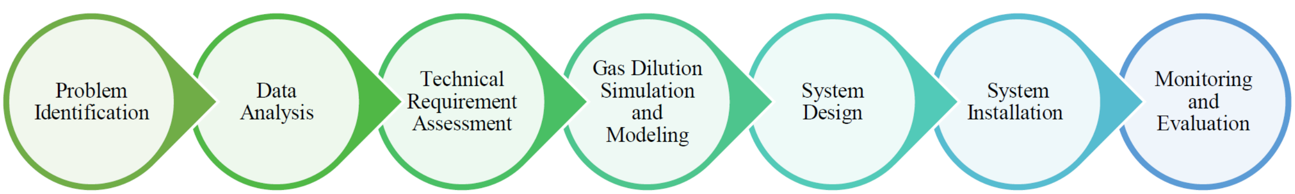

Figure 8—Structured Engineering Workflow

The below chart (Figure 9) is showing the results of example well simulation where IGLS is installed down to 1000 ft assuming 1 ¾" insert string e.g. CT in 4 ½" completion. The quick analysis showing positive sign of H_2S dilution. The lab testing with sour and sweet gas samples are required prior to its field application to fully understand the mixing efficiency under planned operating condition and investigate flow assurance challenges if any. The length of insert string (IGLS) can be longer or set deeper as required and optimized with actual well / reservoir data set, desire H_2S concentration post dilution with available sweet gas, surface operating condition and lab testing results of both sweet & sour gas samples.

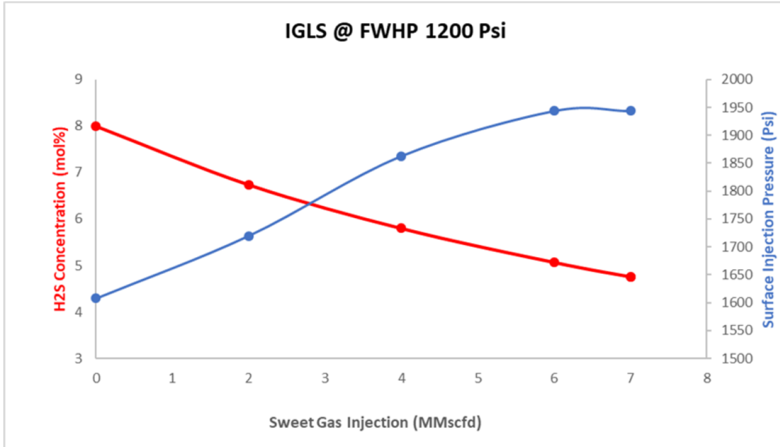

Figure 9—Modelling Results

Technology Added Value

Downhole sweet gas injection, facilitated by inverse gas lift system, presents significant technological advantages over conventional H_2S management methodologies:

- Reduced H_2S content is achieved by the dilution below surface at technically desired achievable depth.
- Enable production from sour wells with minimal effect on downhole and surface equipment.
- Reduced surface processing costs by downhole dilution of H_2S minimizes the reliance on capital-intensive surface H_2S removal units, thereby reducing operating expenditure associated with chemical consumption, waste disposal, and equipment maintenance.
- Mitigated corrosion by lowering the H_2S concentration through controlled downhole injection to mitigates the corrosion rate of wellbore tubular (e.g., production casing, tubing) and surface infrastructure and this extends equipment lifespan and reduces the frequency of costly workovers or replacements.
- Enhanced safety as controlled dilution of H_2S concentrations in the produced fluid stream reduces the risk of exceeding permissible exposure limits for personnel and mitigates potential hazards associated with handling high-concentration sour gas.
- Increased flexibility is achieved as the IGLS system can be dynamically adjusted to accommodate fluctuating well conditions, ensuring sustained and efficient H_2S management over the well's lifecycle.
- Simple to operate, the IGLS does not utilize the original tubing casing annuli (flow through tubing / insert string annuli). The Dual safety valve is an added benefit to well safety while operating the well.

Conclusion

The management of sour wells poses substantial technical, economic, and environmental challenges. Conventional methodologies, including chemical treatments, material selection, and surface processing units, exhibit inherent limitations that can negatively impact the profitability and sustainability of sour gas production. Downhole sweet gas injection presents a viable alternative by mitigating H_2S concentrations directly within the wellbore, thereby reducing the dependency on extensive surface processing infrastructure and minimizing corrosion and safety hazards. The inverse gas lift system further augments this approach, offering a rigless, sustainable, and cost-effective implementation strategy.

Ongoing research and development efforts are imperative to further optimize these technologies and expand the application scope. However, the significant potential benefits afforded by downhole gas injection and inverse gas solutions position them as valuable engineering solutions for unlocking sour gas reservoir potential while minimizing environmental impact and maximizing economic returns.

Acknowledgment

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List of Abbreviations

CRAs: Corrosion-Resistant Alloys
IGLS: Inverse Gas Lift System
DFSV: Dual Flow Safety Valve
DFH: Dual Flow Hanger
SSV: Subsurface Safety Valve
PBR: Polished Bore Receptacle

References

Aborshaid, H.A., AlQasim, H.S., "Bottomhole sweetening of sour gas wells", US Patent 11,970,927 B1, 2022.

Barsoum, B. N., Faramoy, S., and M. M. Zekry. "Diluting Acid Gases With Sweet Gases in Gas Lift Systems: Assessment, Challenges, and Recommendations." Paper presented at the SPE Annual Technical Conference and Exhibition, Dallas, Texas, October 2005. doi: https://doi.org/10.2118/95714-MS

Ahmad Mubashir, Mubashir, Zain Yousfi, Fawad, M. Albadi, Mohamed, A. Baslaib, Mohamed, Al Jeelani, Omar, Salem Aladsani, Amna, Alhouqani, Shamsa, Al Qallabi, Salah, Bankole, Ibukun, Al Jaberi, Fawzi, Dutta, Ashim, El Shahat, Ayman, Alejandro Aranda, Jose, Jibreel, Rami, Matar Ali, Ghada, Agarwal, Anubhav, Fisher Pangestu, Yohannes, Kumar, Rahul, Isiaka Jaiyeola, Abiodun, Yugay, Andrey, Pimenta, Gervasio, Masoud, Rashad, V. Deshmukh, Rohit, Al Shehhi, Hessa, Al Belooshi, Fares, Pendyala, Viswasri, and Chandra Mandal. "Sour Gas Well Testing Challenges-A Successful Case Study." Paper presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, November 2019. doi: https://doi.org/10.2118/197482-MS

Guo, Hongzhi, Li, Huigeng, Weng, Zhiyong, Tao, Ke, and Dong Wang. "Production Enhancement in Deep Sour Gas Field: A Case Study in China." Paper presented at the SPE Russian Petroleum Technology Conference, Virtual, October 2020. doi: https://doi.org/10.2118/202012-MS

Sturman, P.J., Goeres, D.M., and M.A. Winters. "Control of Hydrogen Sulfide in Oil and Gas Wells With Nitrite Injection." Paper presented at the SPE Annual Technical Conference and Exhibition, Houston, Texas, October 1999. doi: https://doi.org/10.2118/56772-MS

Wilhelm, S. Mark, and Russell D. Kane. "Selection of Materials for Sour Service in Petroleum Production." *J Pet Technol* **38** (1986): 1051–1061. doi: https://doi.org/10.2118/16040-PA

Rufat Babayev, Bjorn Sirevaag, Kjell Ivar Grindheim, Scott Kerr, and Britt Sliper, Statoil ASA. "Coiled Tubing Inverted Gas-Lift System Installation on the Norwegian Continental Shelf" SPE-173679-MS

Al-Ghanem, F., Al-Jabri, S., Abdul Hameed, M.R., Al-Saeed, M., Al-Otaibi, M., 2018. Cutting-Edge Solutions for Sour Gas Treatment- A Retrospective Study by Kuwait Oil Company.

Aniekan, O.A., Basil, O.C., Imaobong, B.P., Eelah, M., Ihuoma, G., Conrad, I., Jonathan, M., Onajite, T., Gogo, O., 2023. Reservoir Souring Assessment and Application of Design and Engineering Practice to Successfully Predict and Mitigate H_2S Release in a Produced Water Well Intervention.

Askari, M., Aliofkhazraei, M., Jafari, R., Hamghalam, P., Hajizadeh, A., 2021. Downhole corrosion inhibitors for oil and gas production – a review. *Applied Surface Science Advances* **6**, 100128. https://doi.org/10.1016/j.apsadv.2021.100128

Badrak, R.P., 2018. NACE MR0175/ISO 15156: Update on Current Document and Where are We Going?

Chakhmakhchev, A., 2024. Case Study: Understanding H_2S Occurrence: Bakken Production Using Sulfur Isotopes. *Journal of Petroleum Technology* **76**, 61–64. https://doi.org/10.2118/1024-0061-JPT

Chakhmakhchev, A., Kurz, M.D., McRae, T.A., Yu, X., Abarghani, A., Schmidt, D.D., Azzolina, N.A., Kurz, B.A., Sorensen, J.A., 2024. Investigating H_2S Occurrence in Bakken Oil-Producing Wells Using Sulfur Isotopes in Gas, Water, and Rock Samples.

Chen, T., Wang, Q., Chang, F., Aljeaban, N., Alnoaimi, K., 2020. Recent Development and Remaining Challenges of Iron Sulfide Scale Mitigation in Sour-Gas Wells. *SPE Production & Operations* **35**, 0979–0986. https://doi.org/10.2118/199365-PA

Gamal, H., Lawal, A., Alabdulqader, A., Gaw, H., 2025. Enhancing Profitability Through Technological Innovation: Inverse Gas Lift System as Game-Changer for Well Services.

Hofmann, A., van Strien, W., Malekzadeh, R., 2017. Improving the Efficiency of H_2S Mitigation in Middle East Oil and Gas Fields.

Julian, J.Y., Jackson, J.C., White, T.M., 2014. A History of Gas Lift Valve and Gas Lift Mandrel Damage and Subsequent Retrofit Gas Lift Straddle Installation in Alaska.

Marks, L., Martin, M.R., 2007. Shell's Canadian Sulphur Experience.

Okorocha, I.T., Chinwuko, C.E., Mgbemena, C.E., Mgbemena, C.O., 2020. Gas lift optimization in the oil and gas production process: a review of production challenges and optimization strategies. *International Journal of Industrial Optimization* **1**, 61–70. https://doi.org/10.12928/ijio.v1i2.2470

Siddiqui, M.I., Baber, S., Saleem, W.A., Jafri, M.O., Hafeez, Q., 2013. Industry Practices of Sour Gas Management by Reinjection: Benefits, Methodologies, Economic Evaluation and Case Studies.

Wang, Q., Al-Ajwad, H., Shen, S., Al-Salim, Y., Chen, T., Leal, J., 2019. Sour Corrosion Products Formed in High H_2S Gas Wells.